



9A

PERIODICITY & IONIC BONDS

UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310

LEARNING OBJECTIVES

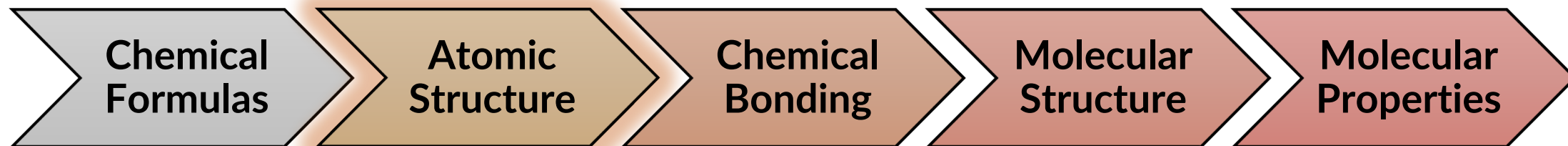
CHAPTER 9

1. Relate each electron configuration and position in the periodic table to the number of valence electrons for an element.
2. Apply Slater's rules to determine effective nuclear charge (Z_{eff}).
3. Predict and rationalize the periodic trends of atomic and ionic radius based on the concept of effective nuclear charge (Z_{eff}).
4. Predict the relative ionization energies of different elements from their positions in the periodic table.
5. Predict the relative electron affinities of different elements from their positions in the periodic table.
6. Determine the charges of common monatomic ions based on the positions of their elements on the periodic table and electron configuration.
7. Describe how electrostatic attractive forces allow ionic compounds to exist as very large, three-dimensional lattices.
8. Describe, apply, and rationalize the periodic trends in lattice energy.

Where did we start and where are we going?

Chemistry in Context

What is our overall goal?



Valence Electrons

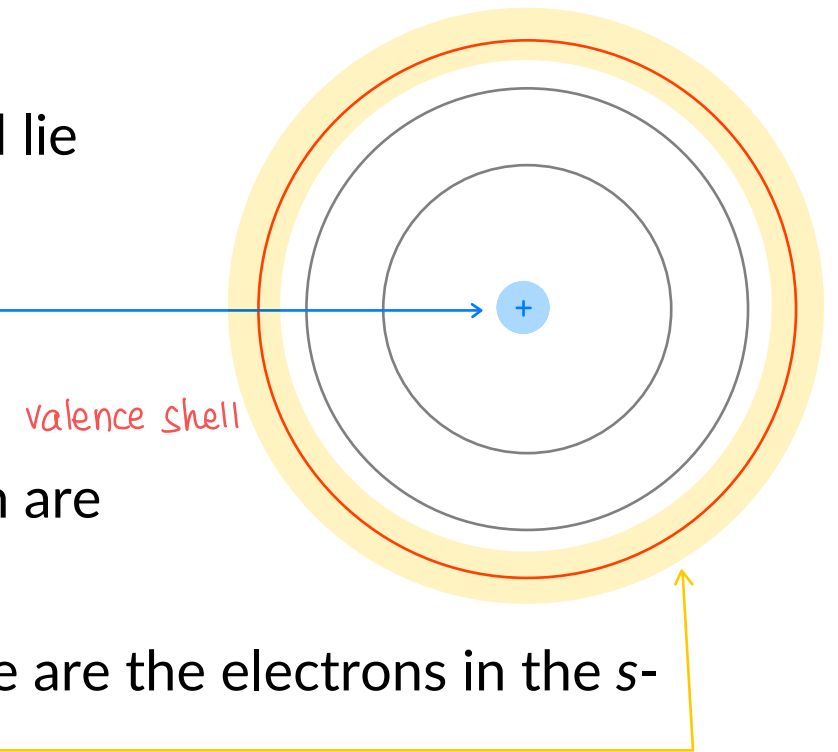
9.1

Core electrons

- The “inner” electrons, which are lower in energy and lie closer to the nucleus.

Valence electrons

- The “outermost” (highest in energy) electrons, which are farther from the nucleus.
- For main group elements (*s*-block and *p*-block), these are the electrons in the *s*- and *p*-orbitals of the highest *n*.
- Valence electrons are involved in bonding, so elements with the same valence electron configurations have similar chemical properties, reactivities, and bonding.



Valence Electrons

9.1

With the exception of helium, the number of valence electrons is equal to the “A” group number from the periodic table.

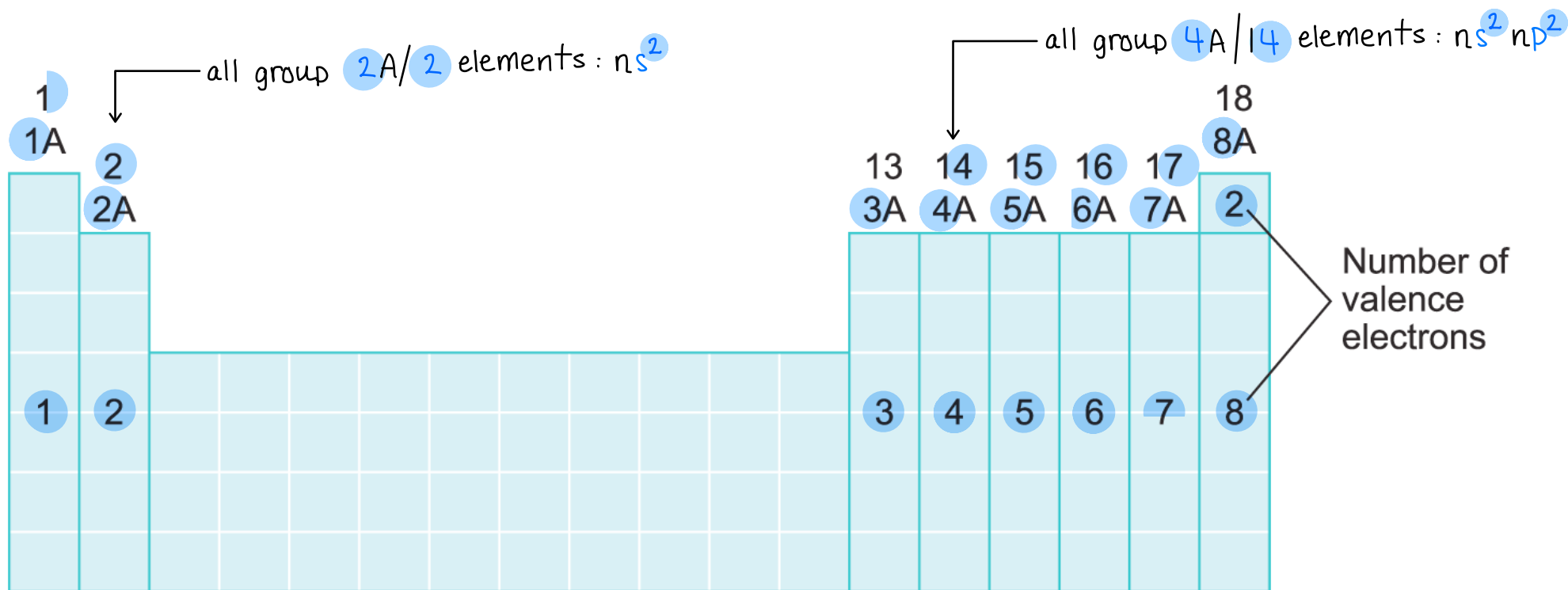


FIGURE 9.1

Atomic and Ionic Sizes

9.2

The sizes of atoms and ions are determined by:

- Electronic structures and electron configuration

of protons , # of e^-

- The interactions between the oppositely charged subatomic particles dictated by electrostatic principles.

positively charged nucleus

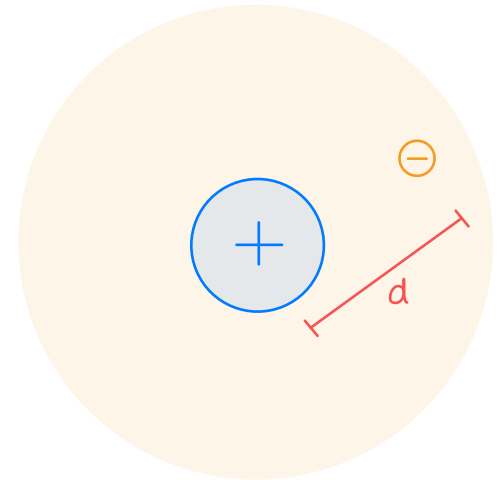
negatively charged electrons

interaction strength increases as charges increase or distance decreases

E_{el}

$q_+ q_-$

d



$$E_{el} = k \frac{q_+ \cdot q_-}{d}$$

Atomic and Ionic Sizes

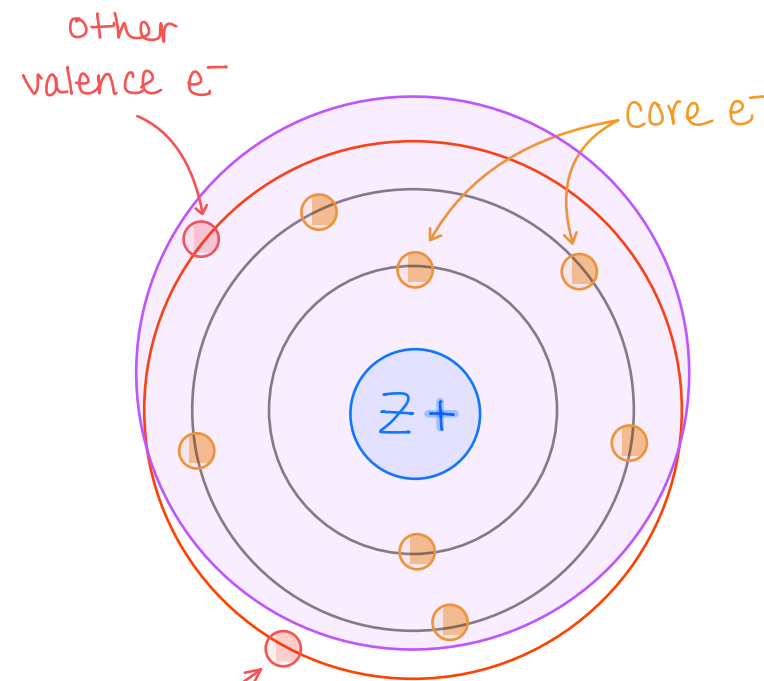
9.2

Nuclear charge (Z)

- The positive charge from protons in the nucleus.

Effective nuclear charge (Z_{eff})

- The net positive charge from the nucleus that an electron experiences.
- The value of $Z_{\text{eff}} < Z$ due to SHIELDING (S).
- The Z_{eff} is calculated for *each electron*, but often the last valence electron.



$$Z_{\text{eff}} = Z - S$$

shielding provided by core e^- and other valence e^-
nuclear charge ($z+$), atomic #, # of protons

Atomic and Ionic Sizes

9.2

Slater's rules to determine the shielding constant (S)

For a given *valence electron in an atom*, split the electrons into two groups:

- 1) Other valence electrons: $(\# \text{ of valence } e^-) - 1$
- 2) *Core electrons*: e^- in orbitals of *lower n* values

$$Z_{\text{eff}} = Z - S$$

$$S = (\# \text{ of core } e^-) \cdot (0.85) + (\# \text{ of other valence } e^-) \cdot (0.35)$$

Element	Electron configuration	Z	# of core e^-	# of other valence e^-	S	Z_{eff}
Na	$1s^2 2s^2 2p^6 3s^1$	11	10	0	$S = (10)(0.85) + (0)(0.35) = 8.5$	$Z_{\text{eff}} = 11 - 8.5 = 2.5$
Mg	$1s^2 2s^2 2p^6 3s^2$	12	10	1	$S = (10)(0.85) + (1)(0.35) = 8.85$	$Z_{\text{eff}} = 12 - 8.85 = 3.15$

Atomic and Ionic Sizes

9.2

Atomic radius

- Increases down a group/column due to size of n
- Decreases from left to right across a period/row due to Z_{eff}

For $n = 3$ row/period:

Element	Z	Z_{eff}
Na	11	2.50
Mg	12	3.15
Al	13	3.80
Si	14	4.45
P	15	5.10
S	16	5.75
Cl	17	6.40

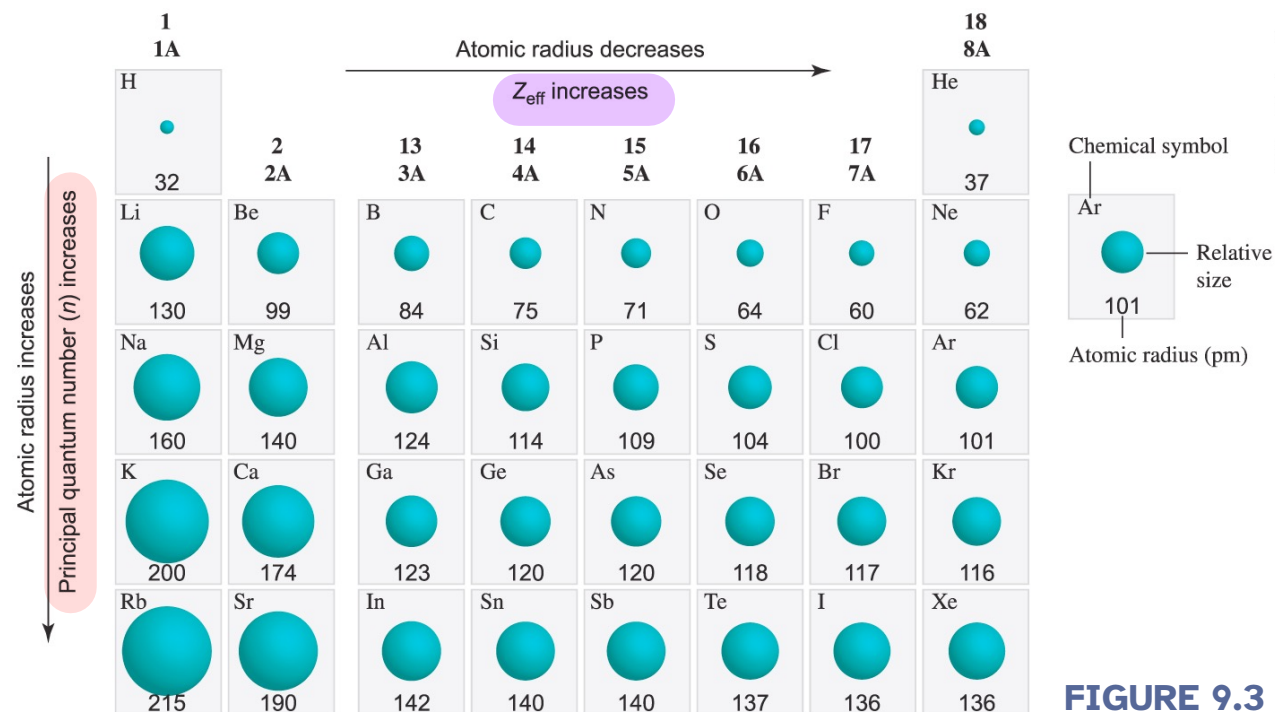


FIGURE 9.3

EXAMPLE PROBLEM

Learning Objective(s): 9.3

Arrange these atoms in order of *increasing radius*: N, O, Na, P.

r decreases left-to-right $O < N$
 $P < Na$

r increases top-to-bottom $N < P$

radius: $O < N < P < Na$

1 H Hydrogen 1.008																	2 He Helium 4.003				
3 Li Lithium 6.941	4 Be Beryllium 9.012															5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 19.00	10 Ne Neon 20.18
11 Na Sodium 22.99	12 Mg Magnesium 24.31															13 Al Aluminum 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97	16 S Sulfur 32.07	17 Cl Chlorine 35.45	18 Ar Argon 39.95
19 K Potassium 39.10	20 Ca Calcium 40.08	21 Sc Scandium 44.96	22 Ti Titanium 47.88	23 V Vanadium 50.94	24 Cr Chromium 52.00	25 Mn Manganese 54.94	26 Fe Iron 55.85	27 Co Cobalt 58.93	28 Ni Nickel 58.69	29 Cu Copper 63.55	30 Zn Zinc 65.38	31 Ga Gallium 69.72	32 Ge Germanium 72.59	33 As Arsenic 74.92	34 Se Selenium 78.96	35 Br Bromine 79.90	36 Kr Krypton 83.80				
37 Rb Rubidium 85.47	38 Sr Strontium 87.62	39 Y Yttrium 88.91	40 Zr Zirconium 91.22	41 Nb Niobium 92.91	42 Mo Molybdenum 95.94	43 Tc Technetium (98)	44 Ru Ruthenium 101.1	45 Rh Rhodium 102.9	46 Pd Palladium 106.4	47 Ag Silver 107.9	48 Cd Cadmium 112.4	49 In Indium 114.8	50 Sn Tin 118.7	51 Sb Antimony 121.8	52 Te Tellurium 127.6	53 I Iodine 126.9	54 Xe Xenon 131.3				
55 Cs Cesium 132.90	56 Ba Barium 137.3	57 La Lanthanum 138.9	72 Hf Hafnium 178.5	73 Ta Tantalum 180.9	74 W Tungsten 183.9	75 Re Rhenium 186.2	76 Os Osmium 190.2	77 Ir Iridium 192.2	78 Pt Platinum 195.1	79 Au Gold 197.0	80 Hg Mercury 200.6	81 Tl Thallium 204.4	82 Pb Lead 207.2	83 Bi Bismuth 209.0	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)				
87 Fr Francium (223)	88 Ra Radium (226)	89 Ac Actinium (227)	104 Rf Rutherfordium (261)	105 Db Dubnium (268)	106 Sg Seaborgium (271)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (276)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (280)	112 Cn Copernicium (285)	113 Nh Nihonium (284)	114 Fl Flerovium (289)	115 Mc Moscovium (288)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)				

IN-CLASS QUESTION 1

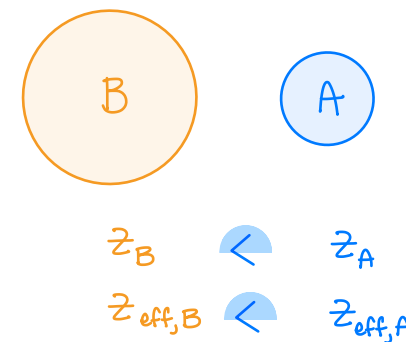
Learning Objective(s): 9.3

Atom A has a smaller atomic radius than atom B. Both atoms are in the ^{same n !} same period.

Which statement *best explains* why this is true?

- ☐ The valence electrons of atom B are more strongly attracted to its nucleus than the valence electrons of atom A are to its nucleus.
- ☐ Atom A has fewer protons than atom B.
- ☒ Z_{eff} for A (for the outermost electron) is greater than Z_{eff} for B.
- ☐ Atom A has more core electrons than atom B.
- ☐ Atom B has more valence electrons than atom A.

atomic size decreases
left to right on PT,
so B before A



Atomic and Ionic Sizes

9.2

All **cations** have *smaller radii* than their corresponding neutral atoms.

All **anions** have *larger radii* than their corresponding neutral atoms.

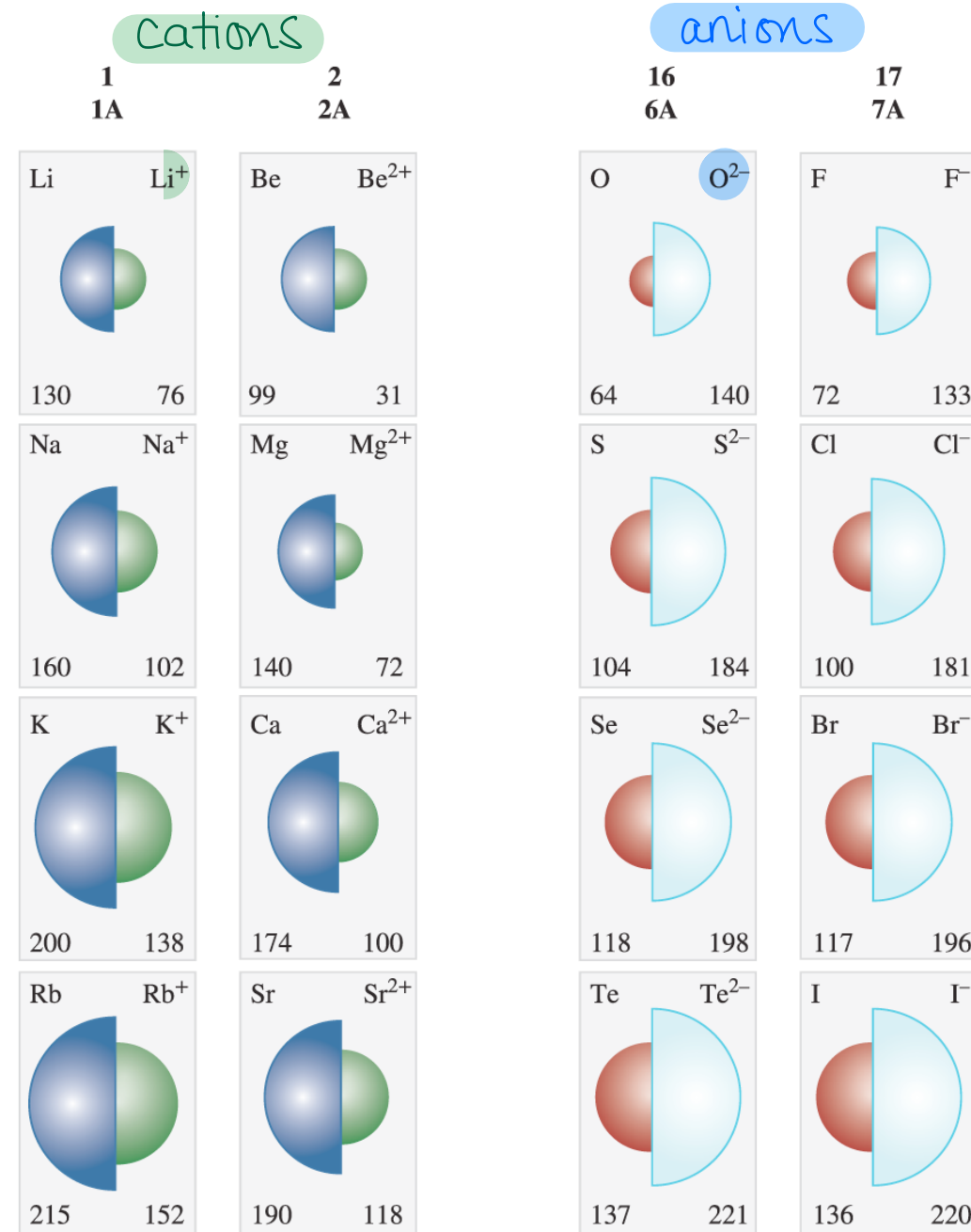
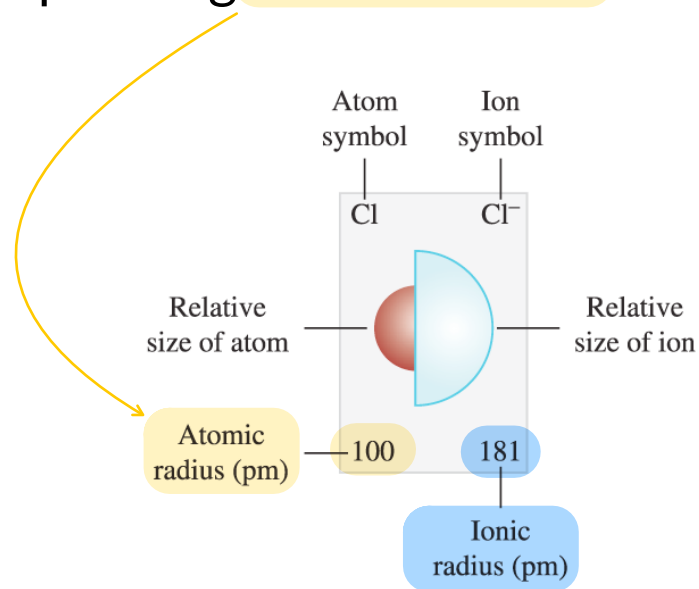
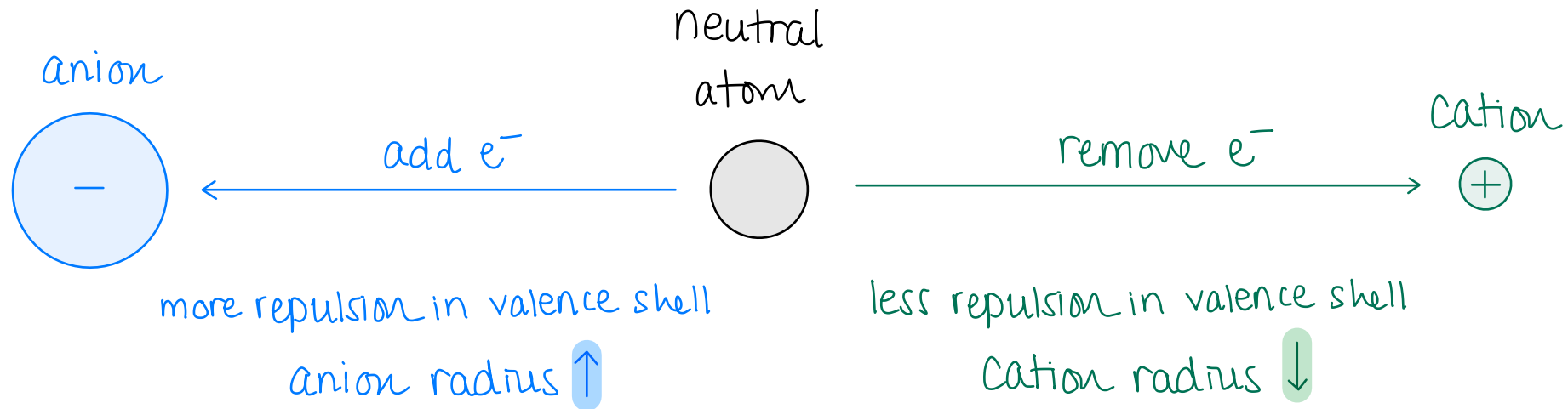


FIGURE 9.5

Atomic and Ionic Sizes

9.2

When ions form, the number of valence electrons changes, but the number of core electrons and protons remains constant.



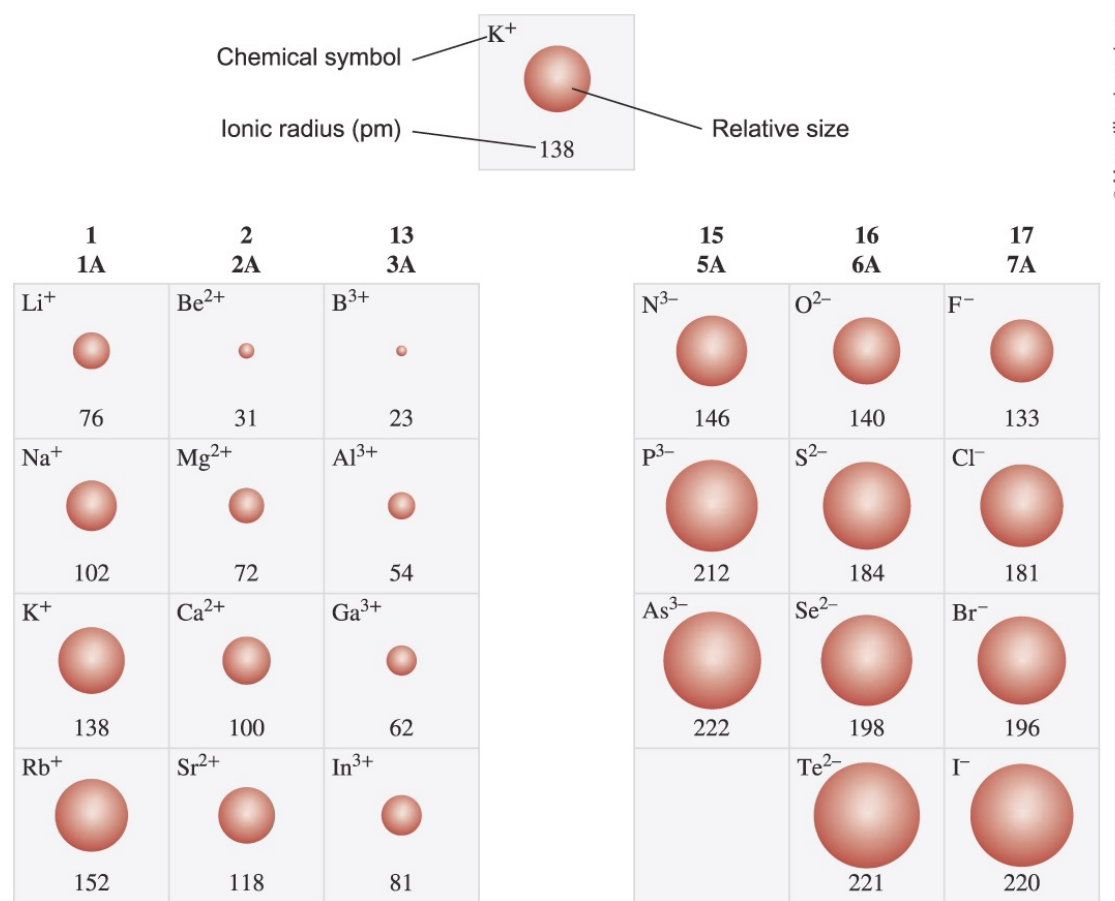
Isoelectronic ions: ions of different elements but same # of electrons ;

need to compare Z and/or Z_{eff}

Atomic and Ionic Sizes

9.2

Cations decrease in size from left to right across the periodic table, as do anions.



note: in a row, most *anions* are larger than *cations*

FIGURE 9.6

IN-CLASS QUESTION 2

Learning Objective(s): 9.3

Predict and arrange the atoms and ions in order of *increasing radius*: O, S²⁻, S, Rb.

r decreases left-to-right O, S < Rb

r increases top-to-bottom O < S



radius: O < S < S²⁻ < Rb

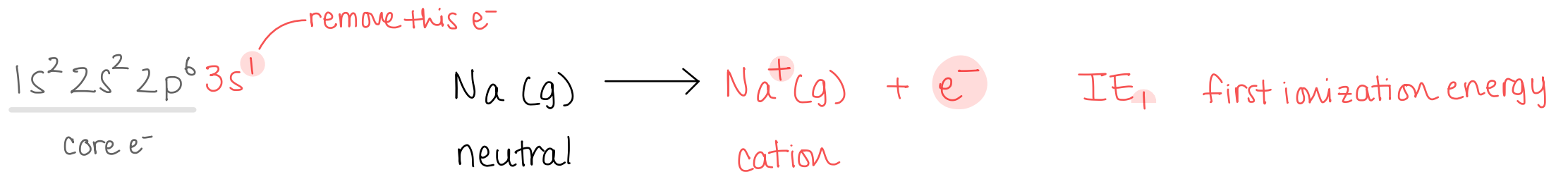
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Ionization Energy and Electron Affinity

9.3

Ionization energy, IE

- The energy required to remove an electron from a neutral gaseous atom to produce a gaseous cation.
- Ionization energies (IE) are always positive.



Ionization Energy and Electron Affinity

9.3

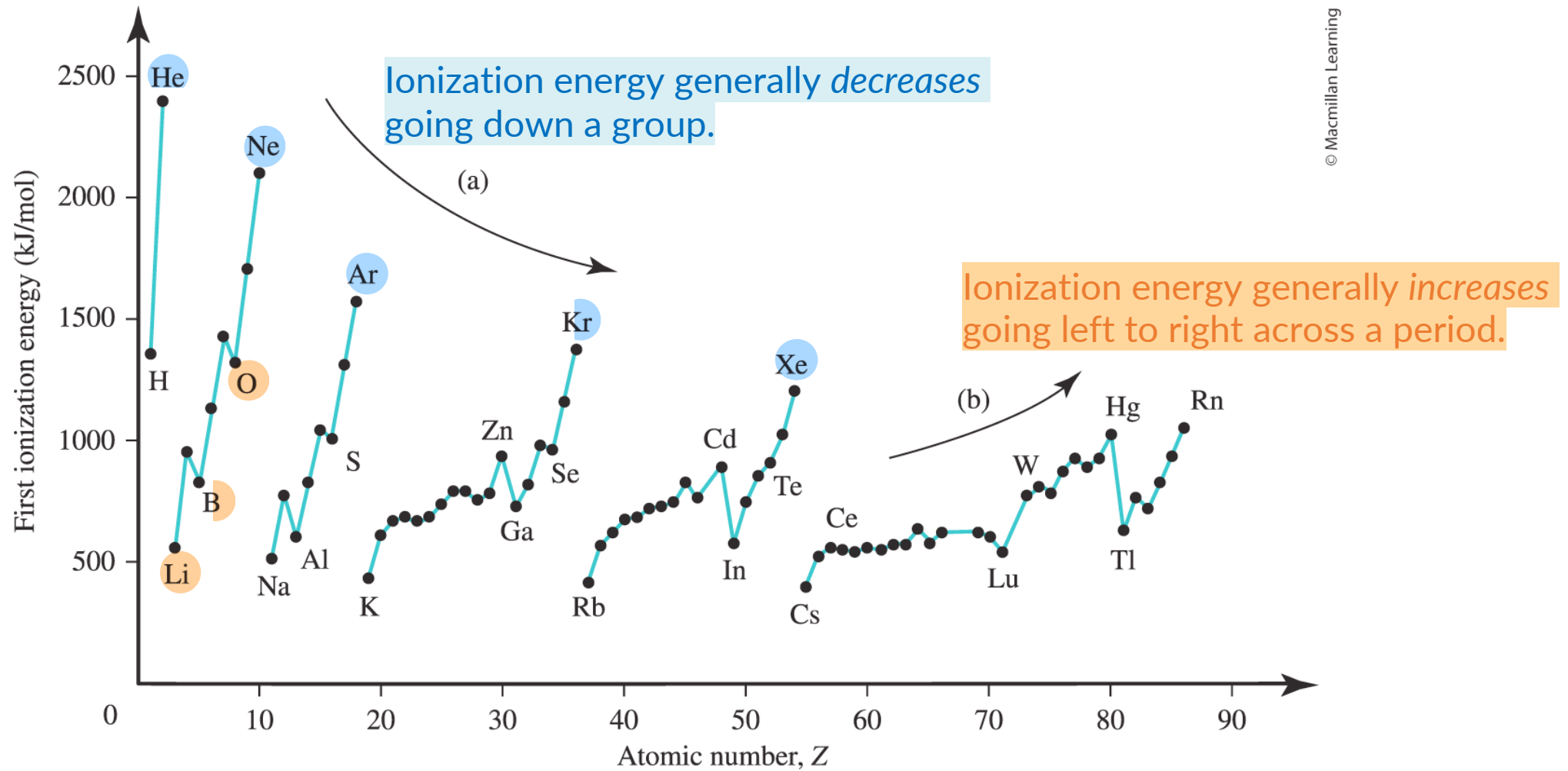
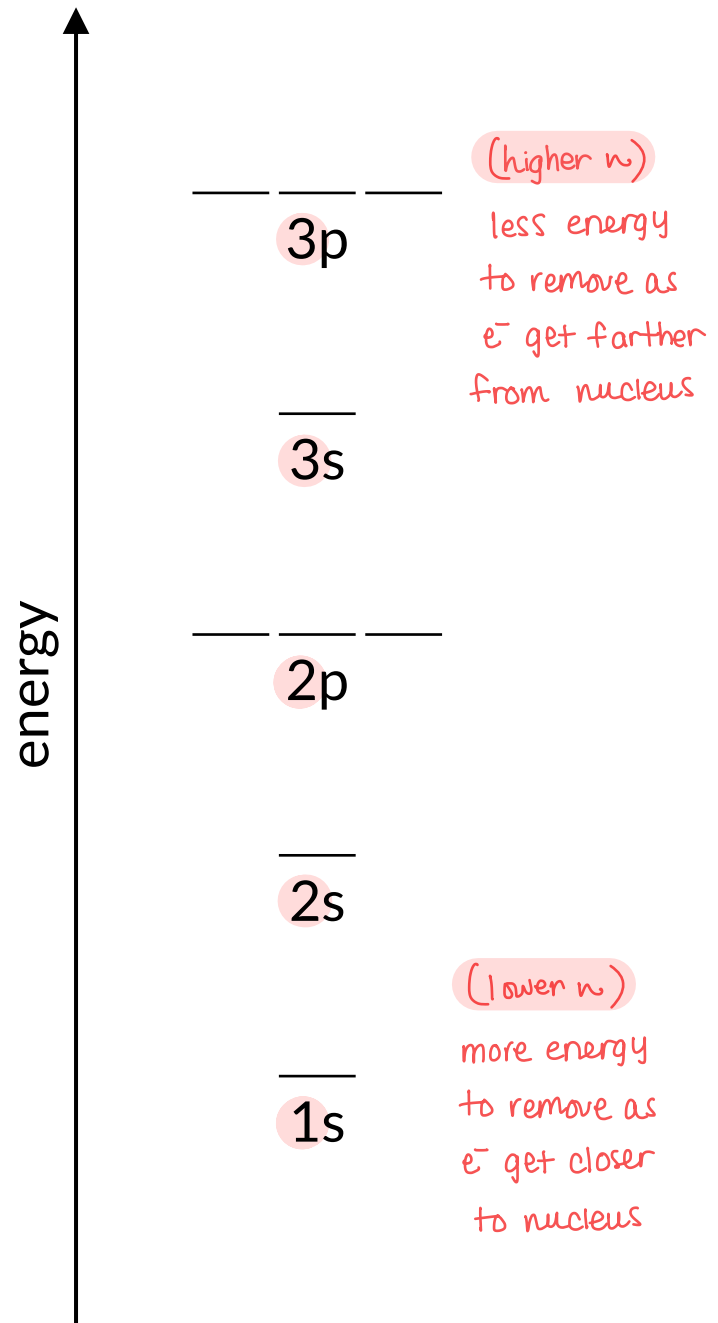


FIGURE 9.7



Ionization Energy and Electron Affinity

9.3

Relative periodic trends in ionization energy and atomic radius.

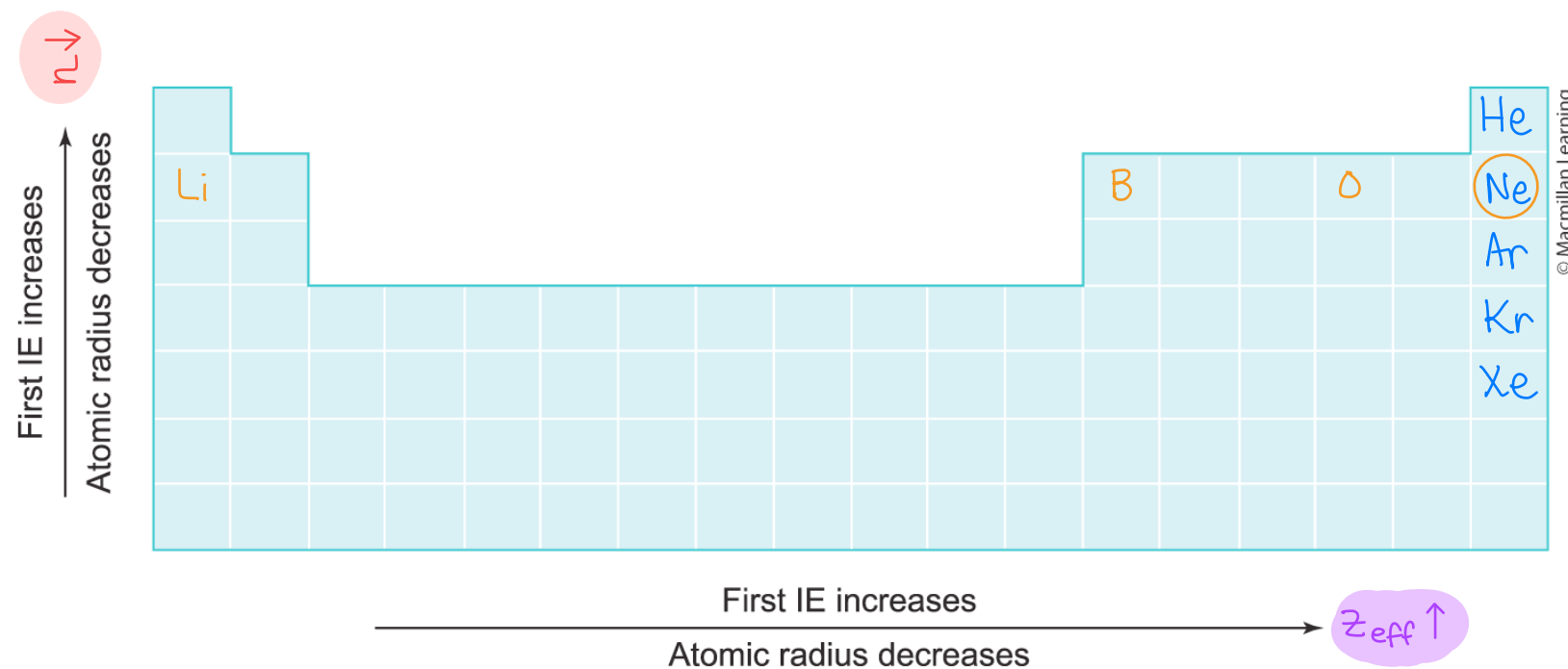


FIGURE 9.8

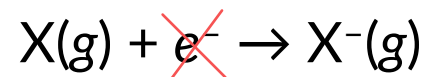
IN-CLASS QUESTION 3

Learning Objective(s): 9.4

Which statement is true regarding the first ionization energy? $A(g) \longrightarrow A^+(g) + e^-$ IE_1

☒ It is the energy ~~released~~ when an electron is removed from a gas phase atom.

☒ The first ionization energy corresponds to the following process:



☒ The first ionization energy for lithium is ~~greater~~ than that for fluorine. $Z_{\text{eff}, F} > Z_{\text{eff}, Li}$

☒ The first ionization energy for lithium should be less than that for sodium because sodium has fewer core electrons.

☒ The first ionization energies for noble gases are very large as a result of their very stable electron configurations.

EXAMPLE PROBLEM

Learning Objective(s): 9.4

Arrange the following atoms in order of *increasing* first ionization energies.

Mg O P Na

IE₁ increases left-to-right Na < Mg < P

IE₁ decreases top-to-bottom P < O

IE₁: Na < Mg < P < O

1 H Hydrogen 1.008																	2 He Helium 4.003				
3 Li Lithium 6.941	4 Be Beryllium 9.012															5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 19.00	10 Ne Neon 20.18
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BONUS QUESTION

Assessed on: Fall 2021, Exam 2

Which atom or ion has the *largest* value of Z_{eff} for the last valence electron?

☐ K

☐ Be

☐ Si

☐ Cl^-

☐ Ca